We suggest that frameworks driven by the Smith acidity scale enable us to predict the space charge properties of oxide-based heterointerfaces. One can therefore expect application of Smith acidity criteria in predicting heterointerface properties to have a very broad impact given that they can be readily applied not only to MIECs, demonstrated by Steinbach et al., but also to semiconducting and even insulating oxides. This makes the approach suitable for application to a much wider temperature range of devices even for microelectronics operating at room temperature. Furthermore, the proposed framework achieved both enhancement and suppression of space charge potential depending on the relative surface acidity. By demonstration and modeling this approach with polycrystalline PCO_{NA} rather than single-crystal samples, it was possible to successfully account for charge transport behavior not only across but also along the heterointerfaces and GBs. This served to provide more realistic insights into acidity-mediated space charge engineered conductivity, with significant implications for practical nanostructured thin film applications.

Summary and outlook

In summary, we have investigated how the PCO_{NA} model platform enables direct observation of orders of magnitude conductivity changes induced at the respective heterointerfaces and GBs, confirming the influence and significance of surface acidity-mediated local space charge potentials. These correlations were firmly established by examining the electronic properties of PCO_{NA} with different infiltrants, insulating substrates, and annealing temperatures. Notably, a single Li-infiltration led to a remarkable three orders of magnitude increase in conductivity. Moreover, our findings challenge the assumption in many studies that insulating substrates do not affect electrical measurements, as their relative acidity can significantly impact measured conductivity at the film substrate interface and upon higher temperature anneals, impact film GB properties as well. Furthermore, our study highlights how the in-diffusion of heterointerface elements at film GBs can also markedly modify the GB transport properties, which needs to be taken into account to fully understand the total changes in conductivity of such nanocrystalline thin films. These features play a direct and critical role in determining the electronic properties of nanostructured functional oxides. This strongly supports the concept of the Smith acidity scale as a predictive descriptor for space charge behavior at heterointerfaces in functional and electrocatalytically active oxide systems. Our understanding of the possible role of supporting substrate in modulating the electronic properties of thin oxide films opens up new opportunities for the modulation of thin film properties without altering their defect chemistry, oxygen stoichiometry, or microstructure.

Methods

Preparation of $Pr_{0.2}Ce_{0.8}O_{2-\delta}$ (PCO20) target source for pulsed laser deposition (PLD)

The PCO20 oxide target was synthesized through a sol-gel method, involving dissolving cerium and praseodymium nitrates with chelating agents in water, adjusting the pH to 9.5, and heating at 80 °C to induce gelation. The gel was dried, fired at 450 °C, calcined at 750 °C to produce PCO20 powder, which was then pelletized and sintered at 1400 °C to form the PLD target.

Fabrication of $Pr_{0.2}Ce_{0.8}O_{2-\delta}$ nanowire arrays (PCO_{NA}) on insulating substrates

PCO_{NA} were fabricated using a combination of the solvent-assisted nanotransfer printing (S-nTP) method and PLD, as illustrated in **Fig. 1a**. Initially, a Si master mold with line-patterned features (line width: 50 nm, line pitch: 150 nm) was prepared via KrF photolithography and reactive ion etching. To facilitate easy detachment of the polymer transfer medium, a layer of polydimethylsiloxane (PDMS, Polymer Source Inc.) was applied to the Si master mold. Subsequently, a solution of polymethylmethacrylate (PMMA, Sigma-Aldrich Inc.) dissolved in a mixed solvent of toluene, acetone, and heptane was spin-cast onto the pre-treated Si master mold to create the polymer transfer medium. The reason for mixing those three solvents is to optimize the surface energy of the layer with respect to the substrate. The PMMA transfer medium was then peeled off from the Si master mold using a polyimide adhesive tape (PI, 3M Inc.), resulting in a reverse morphology of the Si master mold shape. $Pr_{0.2}Ce_{0.8}O_{2-\delta}$ was deposited using PLD with oblique-angle of 80° to guide deposition selectively onto the sections that protrude from the PMMA transfer medium. PLD (Coherent COMPex Pro 205) was performed using a 248 nm KrF excimer laser operating at an energy level of 300 mJ under a frequency of 10 Hz at room temperature. Following deposition, discrete PCO_{NA} were obtained. PCO_{NA} were transfer-printed onto target substrates (Al_2O_3 and MgO, in this study) and the PMMA was removed via washing with toluene or acetone. Due to the limitations imposed by the PMMA transfer medium, high-temperature deposition for improving crystallinity during PLD was not feasible. Thus, the resulting PCO_{NA} on the target substrates were annealed at 700 °C under ambient atmosphere for 2 hours.

Physicochemical characterization of materials

The physical and chemical characterization of PCO_{NA} involved a range of techniques including scanning electron microscopy (SEM), transmission electron microscopy (TEM), X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and inductively coupled plasma